DataCite Corpus - Data Exploration & Descriptive Statistics

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DataCite Corpus - Exploratory Analysis

```
## FUNCTION: exp_summary - argument dat (dataframe)
## Argument dat takes a dataframe. Column names are assumed to
## match the columns provided in the DataCite Corpus as of
## 11 October 2026. Documentation and file downloads available
## at https://zenodo.org/records/16901115

get_dc_summary <- function(filename) {
  ## Returns descriptive statistics of the data file - number
  ## of rows, number & proportion of rows with publication DOIs,
  ## number & proportion of rows with a 'created' date,
  ## number & proportion of rows with publication dates,
  ## number & proportion of rows with a funder listed.

  dat <- read_csv(file = filename, na = c("", "NA", "none"))

  summarylist <- dat |> summarise(
    n = nrow(dat),
    n_doi = str_detect(dat$publication, pattern = "doi") |> sum(na.rm = TRUE),
    n_created_date = sum(1 - is.na(dat$created), na.rm = TRUE),
    n_pub_date = sum(1 - is.na(dat$publishedDate), na.rm = TRUE),
    n_funders = sum(1 - is.na(dat$funders), na.rm = TRUE),
    n_nih = str_detect(dat$funders, pattern = "(NIH)|(National Institutes of Health)") |>
      sum(na.rm = TRUE)
  ) |> unlist()

  return(summarylist)
}


# Function computing descriptive stats

summary_datacite <- function(filenames) {
  cl <- makeCluster(8)
  registerDoParallel(cl)

  summ_dc <- foreach(
    fn = filenames,
    .combine = "rbind",
    .export = "get_dc_summary",
    .packages = "tidyverse"
```

```
  ) %dopar% {
    get_dc_summary(fn)
  }

  stopCluster(cl)

  return(summ_dc)
}
```

```
## TODO: Modify dirpath to current directory
dirpath = "/Users/ashvin/Desktop/biomed-retractions/datacite"

# Gets the names of files to be read int, then calls rw_merge.
files <- list.files(path = paste0(dirpath, "/data"),
                    full.names = TRUE, recursive = TRUE)



summary_dat <- summary_datacite(files) |> as.data.frame()
row.names(summary_dat) <- c(1:nrow(summary_dat))


summary_dat |> summarise(
  N.obs = sum(n),
  N.doi = sum(n_doi),
  Prop.doi = sum(n_doi) / sum(n),
  N.cr.date = sum(n_created_date),
  Prop.cr.date = sum(n_created_date) / sum(n),
  N.funders = sum(n_funders),
  Prop.funders = sum(n_funders) / sum(n),
  N.NIH = sum(n_nih),
  Prop.NIH = sum(n_nih) / sum(n)
) |>
  t() |>
  kable()
```

N.obs	1.069774e+07
N.doi	1.069629e+07
Prop.doi	9.998643e-01
N.cr.date	1.069774e+07
Prop.cr.date	1.000000e+00
N.funders	9.720000e+04
Prop.funders	9.086000e-03
N.NIH	5.202000e+03
Prop.NIH	4.863000e-04

```
df <- read_csv("merged.csv")
```

```
## Rows: 14970 Columns: 21
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (17): data.record.id, data.repository, data.title, data.id, data.citati...
## dbl   (3): pub.record.id, pub.year, pub.retraction.year
```

```
## dttm  (1): data.created.date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
df |> summarise(
  N.obs = nrow(df),
  N.doi = sum(!is.na(data.citation.id)),
  Prop.doi = sum(!is.na(data.citation.id)) / nrow(df),
  N.cr.date = sum(!is.na(data.created.date)),
  Prop.cr.date = sum(!is.na(data.created.date)) / nrow(df),
  N.data.dois = sum(str_detect(data.id, pattern = "10\\.")),
  Prop.data.dois = sum(str_detect(data.id, pattern = "10\\.")) / nrow(df),
  N.pub.date = sum(!is.na(pub.date)),
  N.funders = sum(!is.na(pub.date)) / nrow(df),
  Prop.funders = sum(!is.na(data.funders)) / nrow(df),
  N.NIH = sum(str_detect(data.funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(data.funders.ror, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
              na.rm = TRUE),
  Prop.NIH = sum(str_detect(data.funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(data.funders.ror, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
                na.rm = TRUE) /
    nrow(df)
)  |>
  t() |>
  kable()
```

N.obs	1.49700e+04
N.doi	1.49700e+04
Prop.doi	1.00000e+00
N.cr.date	1.49700e+04
Prop.cr.date	1.00000e+00
N.data.dois	1.27400e+03
Prop.data.dois	8.51035e-02
N.pub.date	1.49700e+04
N.funders	1.00000e+00
Prop.funders	1.00200e-03
N.NIH	4.00000e+00
Prop.NIH	2.67200e-04

Only 1 NIH-funded publication that also appears in RW Database

```
df |> filter(!is.na(data.funders) | !is.na(data.funders.ror)) |>
  select(data.title, pub.title, pub.journal, data.funders, data.funders.ror)
```

```
## # A tibble: 15 x 5
##    data.title                pub.title pub.journal data.funders data.funders.ror
##    <chr>                     <chr>     <chr>       <chr>        <chr>
##  1 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ PEW CHARITA~ PEW CHARITABLE ~
##  2 "<B>RETRACTION:</B>SUPPO~ Dissecti~ Gigascience NATIONAL RE~ NONE NONE; NONE~
##  3 "<B>RETRACTION:</B>SUPPO~ Dissecti~ Gigascience NATIONAL RE~ NONE NONE; NONE~
##  4 "SECOND-GENERATION CITAT~ Effects ~ Chest       ALFRED P. S~ ALFRED P. SLOAN~
##  5 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ UNITED STAT~ NONE NONE; PEW ~
##  6 "HYPOXIA PRECONDITIONED ~ Hypoxia ~ PLoS One    NATIONAL NA~ NATIONAL NATURA~
```

```
##  7 "HINC DATASET FROM: HOME~ Homeopat~ The Oncolo~ INSTITUTE F~ NONE NONE; NONE~
##  8 "HINC DATASET FROM: HOME~ Homeopat~ The Oncolo~ INSTITUTE F~ NONE NONE; NONE~
##  9 "DATA REPOSITORY FOR AUT~ Precise ~ Nature Com~ EU FUNDING ~ NONE NONE; NONE~
## 10 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ UNITED STAT~ NATIONAL INSTIT~
## 11 "EUSPEECH: A NEW DATASET~ Liberals~ PLoS One    HORIZON 2020 NONE NONE
## 12 "LATINO NATIONAL SURVEY ~ Tuning I~ American J~ IRVINE FOUN~ NONE NONE; NONE~
## 13 "PITTSBURGH YOUTH STUDY ~ Chronic ~ Psychology~ UNITED STAT~ NONE NONE; NONE~
## 14 "MONOCENTRIC RETROSPECTI~ Outcomes~ New Microb~ AGENCE NATI~ AGENCE NATIONAL~
## 15 "INVESTIGATION RESULTS O~ Niches o~ Ecology an~ NATIONAL NA~ NATIONAL NATURA~
```

Number of repositories

```
df |> count(data.repository) |> arrange(desc(n))
```

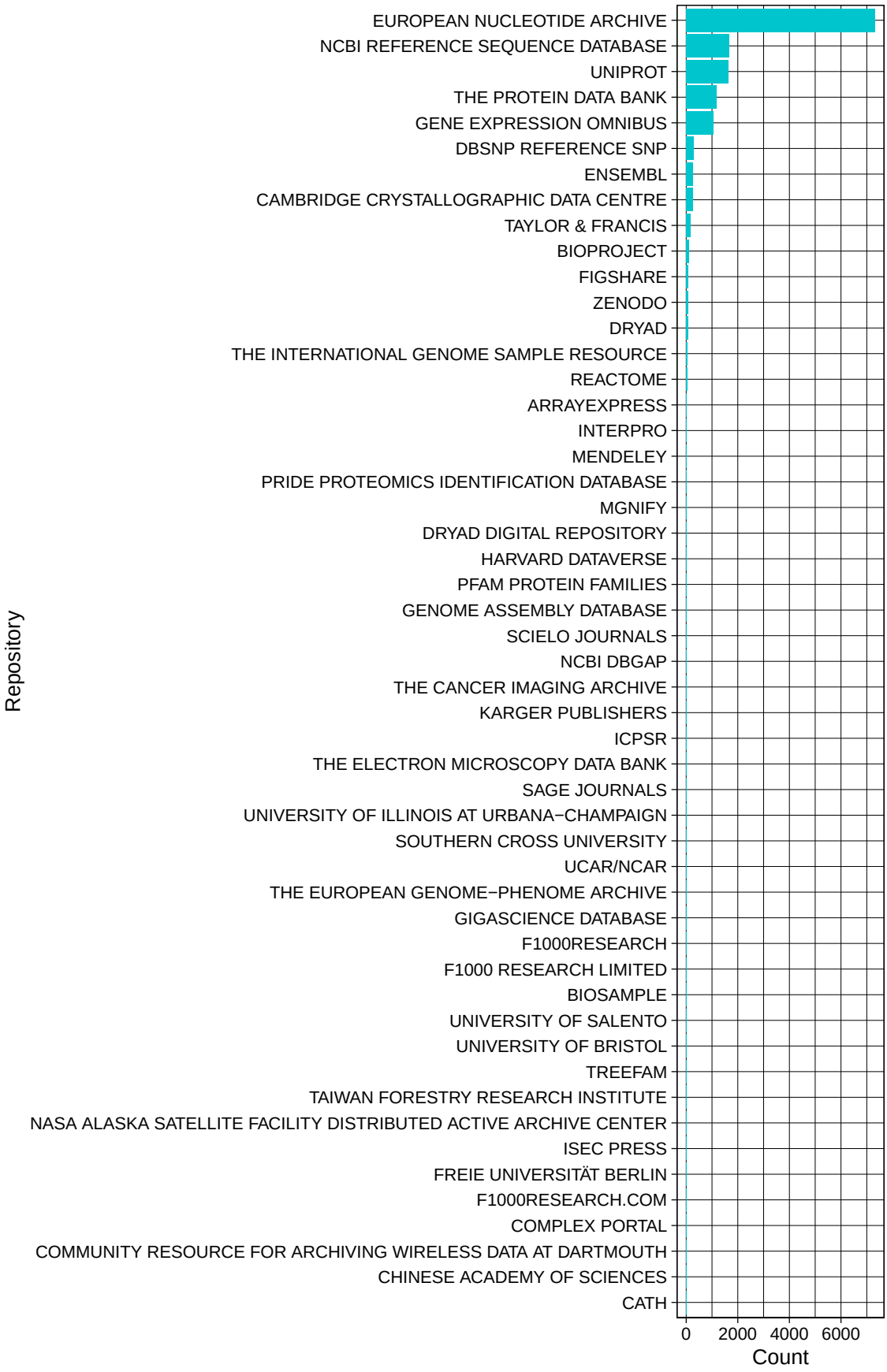
```
## # A tibble: 53 x 2
##    data.repository                          n
##    <chr>                                <int>
##  1 EUROPEAN NUCLEOTIDE ARCHIVE           7316
##  2 NCBI REFERENCE SEQUENCE DATABASE      1655
##  3 UNIPROT                               1641
##  4 THE PROTEIN DATA BANK                 1178
##  5 GENE EXPRESSION OMNIBUS (GEO)         1052
##  6 <NA>                                   519
##  7 DBSNP REFERENCE SNP                    294
##  8 ENSEMBL                                273
##  9 CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE 265
## 10 TAYLOR & FRANCIS                       177
## # i 43 more rows
```

```
df |> filter(str_detect(data.id, pattern = "10\\.")) |>
  count(data.repository) |> arrange(desc(n))
```

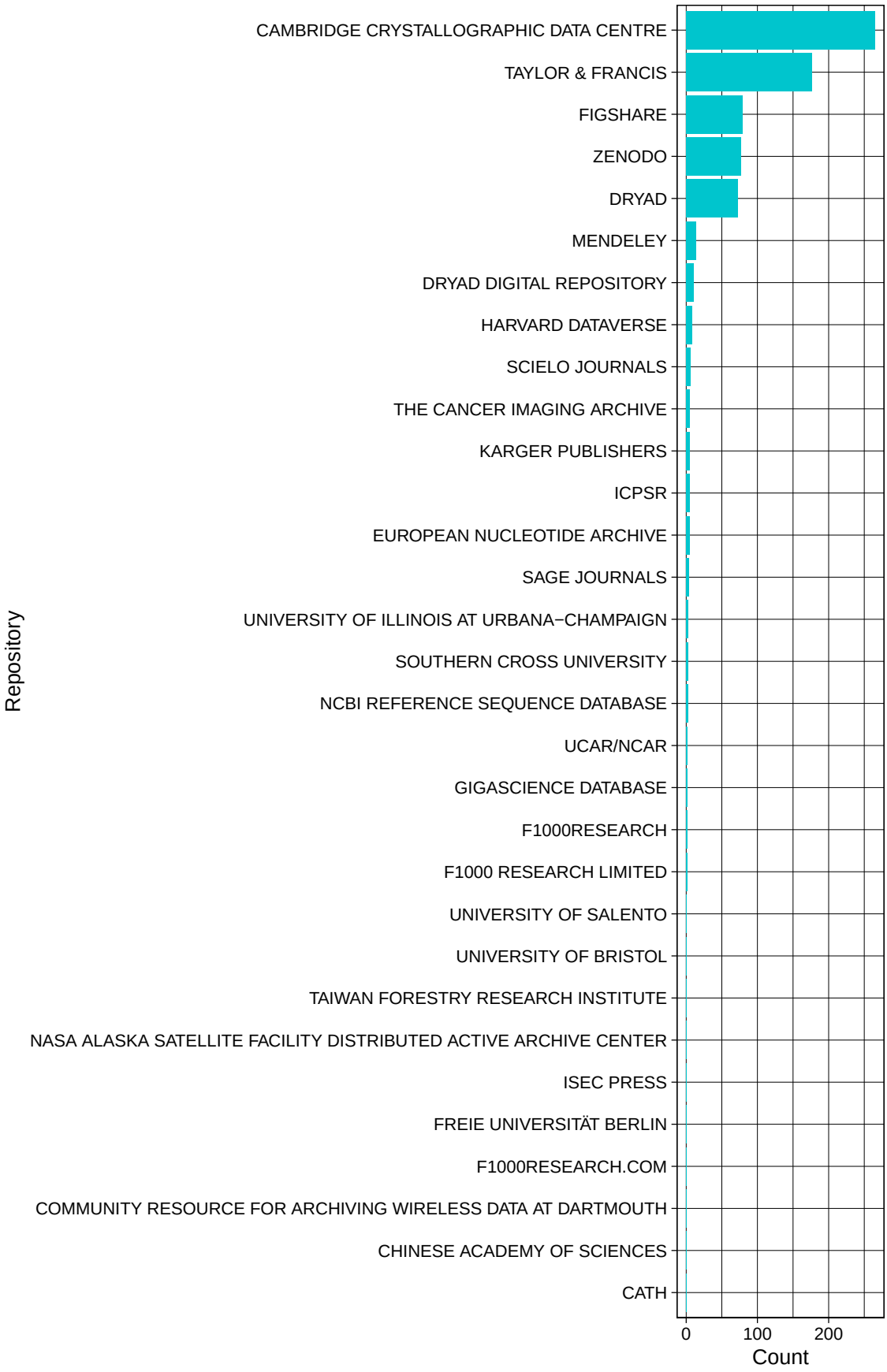
```
## # A tibble: 33 x 2
##    data.repository                          n
##    <chr>                                <int>
##  1 <NA>                                   512
##  2 CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE 265
##  3 TAYLOR & FRANCIS                       177
##  4 FIGSHARE                                79
##  5 ZENODO                                  77
##  6 DRYAD                                   73
##  7 MENDELEY                                14
##  8 DRYAD DIGITAL REPOSITORY                10
##  9 HARVARD DATAVERSE                        8
## 10 SCIELO JOURNALS                          6
## # i 23 more rows
```

```
df |>
  filter(!is.na(data.repository) & data.repository != "UNPUBLISHED") |>
  mutate(
      data.repository.mod = str_remove(data.repository, pattern = "\\s\\(.*\\)")
    ) |>
  mutate(
      data.repository.mod = str_remove(data.repository.mod, pattern = "\\s-\\s.*")
    ) |>
```

```r
  count(data.repository.mod) |>
  ggplot(aes(x = reorder(data.repository.mod, n), y = n)) +
    geom_bar(stat = "identity", fill = "turquoise3") +
    labs(x = "Repository", y = "Count") +
    coord_flip() +
    theme_linedraw()
```


EUROPEAN NUCLEOTIDE ARCHIVE
NCBI REFERENCE SEQUENCE DATABASE
UNIPROT
THE PROTEIN DATA BANK
GENE EXPRESSION OMNIBUS
DBSNP REFERENCE SNP
ENSEMBL
CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE
TAYLOR & FRANCIS
BIOPROJECT
FIGSHARE
ZENODO
DRYAD
THE INTERNATIONAL GENOME SAMPLE RESOURCE
REACTOME
ARRAYEXPRESS
INTERPRO
MENDELEY
PRIDE PROTEOMICS IDENTIFICATION DATABASE
MGNIFY
DRYAD DIGITAL REPOSITORY
HARVARD DATAVERSE
PFAM PROTEIN FAMILIES
GENOME ASSEMBLY DATABASE
SCIELO JOURNALS
NCBI DBGAP
THE CANCER IMAGING ARCHIVE
KARGER PUBLISHERS
ICPSR
THE ELECTRON MICROSCOPY DATA BANK
SAGE JOURNALS
UNIVERSITY OF ILLINOIS AT URBANA−CHAMPAIGN
SOUTHERN CROSS UNIVERSITY
UCAR/NCAR
THE EUROPEAN GENOME−PHENOME ARCHIVE
GIGASCIENCE DATABASE
F1000RESEARCH
F1000 RESEARCH LIMITED
BIOSAMPLE
UNIVERSITY OF SALENTO
UNIVERSITY OF BRISTOL
TREEFAM
TAIWAN FORESTRY RESEARCH INSTITUTE
NASA ALASKA SATELLITE FACILITY DISTRIBUTED ACTIVE ARCHIVE CENTER
ISEC PRESS
FREIE UNIVERSITÄT BERLIN
F1000RESEARCH.COM
COMPLEX PORTAL
COMMUNITY RESOURCE FOR ARCHIVING WIRELESS DATA AT DARTMOUTH
CHINESE ACADEMY OF SCIENCES
CATH
Repository
0
2000
4000
6000
Count

```r
df |> filter(str_detect(data.id, pattern = "10\\.")) |>
  filter(!is.na(data.repository) & data.repository != "UNPUBLISHED") |>
  mutate(
      data.repository.mod = str_remove(data.repository, pattern = "\\s\\(.*\\)")
    ) |>
  mutate(
      data.repository.mod = str_remove(data.repository.mod, pattern = "\\s-\\s.*")
    ) |>
  count(data.repository.mod) |>
  ggplot(aes(x = reorder(data.repository.mod, n), y = n)) +
    geom_bar(stat = "identity", fill = "turquoise3") +
    labs(x = "Repository", y = "Count") +
    coord_flip() +
    theme_linedraw()
```


CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE
TAYLOR & FRANCIS
FIGSHARE
ZENODO
DRYAD
MENDELEY
DRYAD DIGITAL REPOSITORY
HARVARD DATAVERSE
SCIELO JOURNALS
THE CANCER IMAGING ARCHIVE
KARGER PUBLISHERS
ICPSR
EUROPEAN NUCLEOTIDE ARCHIVE
SAGE JOURNALS
UNIVERSITY OF ILLINOIS AT URBANA−CHAMPAIGN
SOUTHERN CROSS UNIVERSITY
NCBI REFERENCE SEQUENCE DATABASE
UCAR/NCAR
GIGASCIENCE DATABASE
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F1000RESEARCH.COM
COMMUNITY RESOURCE FOR ARCHIVING WIRELESS DATA AT DARTMOUTH
CHINESE ACADEMY OF SCIENCES
CATH
Repository
0
100
200
Count

“‘{}